

5. Evaluation metrics

Two conventions govern the whole literature. First, a HIN benchmark normally labels **exactly one node type** — the author in DBLP, the paper in ACM, the movie in IMDB — so “accuracy on DBLP” always means accuracy on that one target type, the other types being context. Second, for *unsupervised* methods the protocol is **embed** → **freeze** → **fit a linear classifier**: the embedding is learned without labels, then held fixed while a logistic regression is trained on a fraction of target nodes, the fraction varied over 20/40/60/80% and mean $\pm$ std reported over repeated splits. HAN, MAGNN and HeCo all report in this form.

5.1 Node classification

With class set $\mathcal{C}$ and per-class counts TP_c, FP_c, FN_c , **Micro-F1** pools the confusion matrix *before* scoring,

$$\begin{aligned} \text{Micro-P} &= \frac{\sum_c TP_c}{\sum_c (TP_c + FP_c)}, & \text{Micro-R} &= \frac{\sum_c TP_c}{\sum_c (TP_c + FN_c)}, \\ \text{Micro-F1} &= \frac{2 \text{Micro-P} \cdot \text{Micro-R}}{\text{Micro-P} + \text{Micro-R}}, \end{aligned}$$

whereas **Macro-F1** scores each class and averages *unweighted*:

$$F1_c = \frac{2 TP_c}{2 TP_c + FP_c + FN_c}, \quad \text{Macro-F1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} F1_c.$$

In the *single-label* multi-class setting every misclassification creates one false positive and one false negative, so $\text{Micro-P} = \text{Micro-R}$ and Micro-F1 equals plain **accuracy**. The pair is always reported together because they answer different questions: Micro-F1 is dominated by frequent classes, Macro-F1 weights a rare class as heavily as a common one, and a large gap between them signals a model that has learned the class prior rather than the graph.

The multi-label caveat. This equivalence **fails** for HGB’s IMDB and Freebase tracks, where a node may carry several labels at once. Prediction is per-label thresholding (sigmoid + binary cross-entropy) rather than an arg max, and the pooled confusion matrix is taken over the flattened node $\times$ label indicator matrix; there $\text{Micro-P} \neq \text{Micro-R}$ in general and Micro-F1 is *not* accuracy. In HGB’s IMDB, 2,684 of the 4,573 labelled movies carry more than one genre, so this is not a corner case — and it is why a Micro-F1 on HGB’s IMDB cannot be compared with one on MAGNN’s single-label IMDB.

Other classification metrics. **Accuracy** is the official metric on OGB (ogbn-mag, MAG240M). **ROC-AUC** is used for binary or strongly imbalanced targets (and by HeCo alongside F1). For the *clustering* probe that HAN, MAGNN, DMGI and HeCo all run — k -means on the frozen embeddings with $k = |\mathcal{C}|$ — the scores are

$$\text{NMI}(Y, \hat{C}) = \frac{2 I(Y; \hat{C})}{H(Y) + H(\hat{C})} \in [0, 1], \quad \text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]},$$

with I mutual information, H entropy and RI the Rand index. Both are 0 in expectation for a chance partition and 1 for a perfect one (ARI may go negative). These test whether the embedding geometry is class-structured with no classifier fitted at all.

5.2 Link prediction

The essential difference from the homogeneous case is that in a HIN link prediction is stated **per relation type**: not “is there an edge” but “is there an edge *of type R*” — predict Author–Paper, or Paper–Venue, or User–Artist. Two consequences follow. (i) *Negatives must be sampled type-consistently*: a negative for an Author–Paper edge must join an author to a paper, never an author to a venue, since a type-violating negative is trivially rejected by any type-aware model and inflates its apparent margin over type-agnostic baselines. HGB samples negatives from 2-hop neighbours at a 1:1 ratio for exactly this reason. (ii) Scores are reported *per relation* and then averaged, so a model can be strong on a dense relation and useless on a sparse one without that showing in a single number.

Let rank_i be the position of the i -th true edge among its candidates, over query set Q :

$$\text{MRR} = \frac{1}{|Q|} \sum_{i \in Q} \frac{1}{\text{rank}_i}, \quad \text{Hits@}K = \frac{1}{|Q|} \sum_{i \in Q} \mathbf{1}[\text{rank}_i \leq K],$$

i.e. MRR is the mean reciprocal rank of the first true item and Hits@ K the fraction of positive edges ranked inside the top K against their sampled negatives ($K \in \{1, 3, 10\}$ conventionally, inherited from the KG literature). **ROC-AUC** equals the probability that a random positive outranks a random negative. **Average precision** (equivalently AUPRC) is

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n,$$

with P_n, R_n precision and recall at the n -th threshold. AP is preferred to ROC-AUC when positives are rare — always true for link prediction on a sparse graph — because ROC-AUC is optimistic there, the enormous true-negative count sitting in its denominator. **Precision@ K / Recall@ K** , and MAE/RMSE for rating prediction, appear in the recommendation-flavoured HIN papers (HERec on Yelp/Douban). HGB’s LP track standardises on **ROC-AUC and MRR**.

5.3 Summary: dataset $\rightarrow$ target $\rightarrow$ task $\rightarrow$ metric

Table 1:

Dataset	Domain	Target	Task	Labels	Metrics normally reported
DBLP	bibliographic	author	NC	4, single	Macro-F1, Micro-F1 (+NMI/ARI)
ACM	bibliographic	paper	NC	3, single	Macro-F1, Micro-F1 (+NMI/ARI)
IMDB (HGB)	movie	movie	NC	5, multi	Macro-F1, Micro-F1
IMDB (MAGNN)	movie	movie	NC	3, single	Macro-F1, Micro-F1
Freebase (HGB)	KG	book	NC	7, single	Macro-F1, Micro-F1
AMiner	bibliographic	author/paper	NC	8 / 4	Macro-F1, Micro-F1, NMI
ogbn-mag	bibliographic	paper	NC	349, single	Accuracy (official)
MAG240M	bibliographic	paper	NC	153, single	Accuracy (official)
LastFM	music	user–artist	LP	—	ROC-AUC, MRR
Amazon	e-commerce	product–product	LP	—	ROC-AUC, MRR
PubMed (HGB)	biomedical	disease–disease	LP	—	ROC-AUC, MRR
Yelp / Douban	review	user–item	LP / rating	—	Precision@K, Recall@K; MAE, RMS
OAG (HGT)	bibliographic	paper–field / venue	LP	—	NDCG, MRR

Closing remark. The single most important methodological point is the one HGB was created to make: because DBLP, ACM and IMDB circulate in mutually incompatible preprocessed versions (Table ??), and because IMDB even changes from single-label to multi-label between versions, numbers copied across papers are frequently not comparable. Any credible evaluation must name the dataset *version*, the split, and whether the protocol is end-to-end supervised or embed-freeze-classify.